

Pull-Back Toy Motion Lab: Student Data Audit

Unit 2: Kinematics & Uniform Acceleration • Data Integrity & Systematic Anomaly Review

125  
TOTAL STUDENTS

86  
AUTHENTIC RUNS (68.8%)

18  
FLAGGED BOGUS (14.4%)

21  
INCOMPLETE / DRAFT (16.8%)

● AUTHENTIC PHYSICAL RUNS: 86 (68.8%)

● INCOMPLETE / DRAFTS: 21 (16.8%)

● FLAGGED / UNPHYSICAL SETS: 18 (14.4%)

1. SECTION PACING & DATA INTEGRITY MATRIX

| CLASS SECTION              | ROSTER | TURNED IN | AUTHENTIC  | FLAGGED BOGUS | INCOMPLETE | PRIMARY ANOMALY DIAGNOSTIC                                                   |
|----------------------------|--------|-----------|------------|---------------|------------|------------------------------------------------------------------------------|
| Period 0 (Honors Physics)  | 14     | 14 (100%) | 14 (100%)  | 0 (0.0%)      | 0          | 100% CLEAN Exemplary slow-mo telemetry (0.75 s – 1.05 s).                    |
| Period 1 (Conceptual Phys) | 10     | 6 (60%)   | 5 (50.0%)  | 2 (20.0%)     | 3          | Raw video frames entered as seconds: [8, 27, 41, 51, 67] (67 s total).       |
| Period 2 (Conceptual Phys) | 13     | 7 (54%)   | 4 (30.8%)  | 4 (30.8%)     | 5          | Pod sharing synthetic [1, 2, 3, 4, 5] integer placeholders (0 accel).        |
| Period 3 (Conceptual Phys) | 14     | 13 (93%)  | 9 (64.3%)  | 4 (28.6%)     | 1          | Group typo: entered 91.00 s for 100 cm mark (took 89.8 s for 20 cm).         |
| Period 4 (Physics Regular) | 31     | 21 (68%)  | 25 (80.6%) | 0 (0.0%)      | 6          | ZERO BOGUS 25 clean accelerating runs; 6 in progress.                        |
| Period 5 (Physics Regular) | 31     | 25 (81%)  | 24 (77.4%) | 4 (12.9%)     | 3          | Distance typed as time (100 s); video frame numbers (79 s); stall.           |
| Period 6 (Physics Regular) | 12     | 8 (67%)   | 5 (41.7%)  | 4 (33.3%)     | 3          | Fabricated constant speed (0.25 s steps), double-tap (0.18 s), frame switch. |

2. MASTER ANOMALY DIRECTORY (18 FLAGGED STUDENTS ACROSS 9 CLUSTERS • ORDERED BY PERIOD)

| PER | FLAGGED STUDENTS & IDS                                                                               | RECORDED TIMESTAMPS (0 TO 100 CM)  | TOTAL   | PATHOLOGY TYPE       | DIAGNOSTIC FINDING                                                                 |
|-----|------------------------------------------------------------------------------------------------------|------------------------------------|---------|----------------------|------------------------------------------------------------------------------------|
| P1  | Destiny Carrasco (396854), Daniela Rivera (431303)                                                   | [0, 8.0, 27.0, 41.0, 51.0, 67.0]   | 67.0 s  | Frame Numbers        | Hundredths/frames entered as seconds (car took 67 s; v = 1.5 cm/s).                |
| P2  | Anthony Baez (389113), Marco Figueroa (397870), Azaleah Buenaflor (398861)                           | [0, 1.00, 2.00, 3.00, 4.00, 5.00]  | 5.00 s  | Placeholders         | Fabricated sequential integers (constant 20 cm/s; zero acceleration).              |
| P2  | Randy Valencia Molina (387527)                                                                       | [0, 1.00, 2.00, 3.00, 4.00, 4.99]  | 4.99 s  | Placeholders         | Copied 1, 2, 3, 4 then typed 4.99 s to bypass app duplicate validator.             |
| P3  | Lesslie Enriquez (390808), Jimena Herrera (430156), Emily Miranda (435778), Marleny Vazquez (441768) | [0, 0.30, 0.36, 0.70, 1.21, 91.00] | 91.0 s  | Shared Video Typo    | Typo on 100 cm mark (car took 89.8 s for 20 cm). Video reads ~1.91 s.              |
| P5  | Joshua Escobar (393188), Alexander Rosas (395643)                                                    | [0, 30.0, 47.0, 60.0, 65.0, 79.0]  | 79.0 s  | Frame Numbers        | Slow-mo video frame counts entered directly as seconds (79 s run).                 |
| P5  | Kevin Velazquez (389920)                                                                             | [0, 20.0, 40.0, 60.0, 80.0, 100.0] | 100.0 s | Distance as Time     | Copied track distance tape markers (20-100 cm) into elapsed time boxes.            |
| P5  | Valerie Cervantes (396976)                                                                           | [0, 0.05, 0.26, 1.29, 6.44, 7.03]  | 7.03 s  | Severe Decel / Stall | Initial speed 400 cm/s; final interval crawled at 3.9 cm/s (car collision/stall).  |
| P6  | Skyler Almaguer (390179), Oscar Bonifacio (392126)                                                   | [0, 0.25, 0.50, 0.75, 1.00, 1.25]  | 1.25 s  | Synthetic Linear     | Identical arithmetic increments (stdev = 0.00 s; zero accel; constant speed).      |
| P6  | Allison Menjivar (392908)                                                                            | [0, 0.08, 0.09, 0.11, 0.14, 0.18]  | 0.18 s  | Teleportation        | 100 cm in 0.18 s (12.4 mph); interval 2 was 0.01 s (45 mph). Stopwatch double-tap. |
| P6  | Bryan Garcia Avalos (396259)                                                                         | [0, 0.92, 41.0, 65.0, 83.0, 120.0] | 120.0 s | Mid-Table Switch     | Entered 0.92 s for 20 cm, then switched to video frame numbers (41 to 120).        |



Action Protocol Guide: Turn to Page 2 for the dedicated, period-by-period teacher intervention protocol with specific student diagnostics, conversational remediation scripts, and pending draft checklists.

PERIOD 0 HONORS 14 / 14 Authentic Submissions (100% Clean) • No in-class remediation required.

Active Follow-ups: 18 Flagged Students • 21 Incomplete Drafts

PERIOD 1 Conceptual Physics

2 Flagged • 3 Incomplete

PROBLEM CITED: VIDEO FRAME NUMBERS AS SECONDS

Entered raw slow-mo frame counts / hundredths [8, 27, 41, 51, 67] as seconds. The car calculates as taking 67.0 s to travel 1.0 m (average speed of 1.5 cm/s).

Flagged Students: Destiny Carrasco (396854), Daniela Rivera Carbajal (431303)

SUPPORT NEEDED / IN-CLASS ACTION:

Have students open their run video in their gallery. Explain that the digits in the scrubber overlay are video frame ticks or hundredths. Guide them to divide entries by 100 or re-read the decimal seconds display (e.g. 0.67 s instead of 67). Updating values immediately restores their velocity calculation to an authentic ~149 cm/s.

PENDING DRAFTS CHECKLIST:

Matthew Rodriguez Garcia (392998), Brianna Lozada (398089), Pablo Galicia (403474).  
Remedy: Pair with bench partners who have recorded slow-mo videos to record track times.

PERIOD 2 Conceptual Physics

4 Flagged • 5 Incomplete

PROBLEM CITED: SYNTHETIC INTEGER PLACEHOLDERS (0 ACCEL)

Entered synthetic integers [0, 1.00, 2.00, 3.00, 4.00, 5.00] (constant 20.0 cm/s; acceleration = 0.0 cm/s<sup>2</sup>). Randy typed 4.99 s at the end to trick the duplicate checker.

Flagged Students: Anthony Baez (389113), Marco Figueroa (397870), Azaleah Buenaflor (398861), Randy Valencia Molina (387527)

SUPPORT NEEDED / IN-CLASS ACTION:

Invalidate/reset drafts in teacher panel. Explain that pull-back cars accelerate from rest and cannot take exactly 1.000 s per 20 cm split. Require pod to present phone slow-mo video showing the car on the 100 cm tape before re-entering authentic decimal splits.

PENDING DRAFTS CHECKLIST:

Anthony Quinones Sanchez (395193), Aiden Gutierrez-Torres (398816), Kaylee Fabian Martinez (417149), Ricky Marquez (421804), Jacqueline Cabrera (425747).  
Remedy: Check lab stations; provide working pull-back cars and stopwatch timers.

PERIOD 3 Conceptual Physics

4 Flagged • 1 Incomplete

PROBLEM CITED: SHARED TYPO ON 100 CM MARK (91.00 S)

Splits 1–4 are realistic (0.30 s to 1.21 s), but final mark was entered as 91.00 s instead of ~1.91 s. Interval 4 → 5 implies the car took 89.8 s for 20 cm (0.22 cm/s crawl).

Flagged Pod: Lesslie Enriquez Rodriguez (390808), Jimena Herrera Morales (430156), Emily Miranda Silio (435778), Marleny Vazquez Ibarra (441768)

SUPPORT NEEDED: 60-SECOND IN-CLASS FIX

Have the 4 students open their webapp. Ask: "Did your car accelerate through 80 cm in 1.2 s, and then crawl for a minute and a half to go the last 20 cm?" Check their phone video at 100 cm: it reads 01:91. Have them change 91.00 to 1.91. The graph and velocities instantly snap to an acceleration curve!

PENDING DRAFTS CHECKLIST:

Arturo Palos (396517) — Remind Arturo to copy his group's corrected timestamps and submit.

PERIOD 4 Physics Regular

0 Flagged • 6 Drafts Pending

STATUS: 100% AUTHENTIC DATA — ZERO BOGUS SETS!

All 25 completed submissions show genuine physical acceleration curves (average speeds 85–140 cm/s). Zero synthetic fabrication or unit errors detected across the section!

PRIORITY ACTION: 4 VALID RUNS NEED FINAL SUBMISSION:

4 students have already entered authentic, valid timestamps into their draft table but haven't clicked submit:

- Sebastian Reynoso (391714): [0, 0.39, 0.56, 0.71, 0.80, 0.95]
- Jacob Sanchez (393031): [0, 0.31, 0.48, 0.58, 0.68, 0.79]
- Miguel Alcala Laurean (395419): [0, 0.38, 0.56, 0.71, 0.83, 0.95]
- Wychane Randle (438294): [0, 0.41, 0.62, 0.79, 0.93, 1.23]

Guide them to compute average/final velocity, acceleration, and hit "Submit Lab".

BLANK DRAFTS CHECKLIST (6 STUDENTS):

Joshua Sheldon (391960), Dantae Valdez (392421), Sabrina Anaya Trinidad (396311), Jazmin Panzo Garcia (396323), Bryssa Diaz (396686), Jose Guzman (397139). Prompt them to test & submit.

PERIOD 5 Physics Regular

4 Flagged • 3 Incomplete

PROBLEMS CITED: 3 DISTINCT ANOMALIES

- Frames: Escobar & Rosas entered [30, 47, 60, 65, 79] (79 s run).
- Distance as Time: Velazquez typed [20, 40, 60, 80, 100] (100 s run).
- Severe Stall: Cervantes entered 0.05 s then 7.03 s (car died/crashed).

Flagged Students: Joshua Escobar (393188), Alexander Rosas (395643), Kevin Velazquez (389920), Valerie Cervantes (396976)

SUPPORT NEEDED / IN-CLASS ACTION:

- Escobar & Rosas: Coach to divide frame entries by 100 or read seconds scrubber.
- Velazquez: Explain 20-100 are cm distances; re-enter stopwatch elapsed seconds.
- Cervantes: Check if car hit a ruler edge; re-film one clean 100 cm run on tile.

PENDING DRAFTS CHECKLIST:

Valid times entered, need submit: Britany Cruz Ramirez (437688), Yaretzy Solis Ruiz (445225). Blank drafts: Omar Olivares (391057), Kimberly Vargas (391727), Tianna Calderon (395490).

PERIOD 6 Physics Regular

4 Flagged • 3 Incomplete

PROBLEMS CITED: 3 DISTINCT ANOMALIES

- Synthetic Linear: Almaguer & Bonifacio entered [0.25, 0.50, 0.75, 1.00, 1.25].
- Double-Tap: Menjivar total run = 0.18 s (interval 2 was 0.01 s = 45 mph!).
- Frame Switch: Garcia Avalos entered 0.92 s, then switched to frames 41-120.

Flagged Students: Skyler Almaguer (390179), Oscar Bonifacio (392126), Allison Menjivar (392908), Bryan Garcia Avalos (396259)

SUPPORT NEEDED / IN-CLASS ACTION:

- Almaguer & Bonifacio: Explain accelerating intervals shorten; re-measure from video.
- Menjivar: Explain reaction time limits; use video scrubber to get accurate split times.
- Garcia Avalos: Convert frames 41-120 into decimal seconds for uniform units.

PENDING DRAFTS CHECKLIST:

Finley De Vaul (420924) (partial times). Blank drafts: Josiah Cook (396166), Arely Acevedo Saavedra (412208).